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1  /*
2  --> Java Hashtable class:
3
4  • Java Hashtable class implements a hashtable, which maps keys to
  values.
5  • It inherits Dictionary class and implements the Map interface.
6
7  NOTE:
8
9  • A Hashtable is an array of a list. Each list is known as a bucket.
10 • The position of the bucket is identified by calling the hashCode()
  method.
11 • A Hashtable contains values based on the key.
12 • Java Hashtable class contains unique elements.
13 • Java Hashtable class doesn't allow null key or value.
14 • Java Hashtable class is synchronized.
15 • The initial default capacity of Hashtable class is 11.
16
17 * Constructors:
18
19 1. Hashtable ()
20 2. Hashtable (int initialcapacity)
21 3. Hashtable (int initialcapacity, float loadfactor)
22 4 .Hashtable (Map t)
23
24  */
25
26 import java.util.*;
27 import java.util.Collection;
28 import java.util.Hashtable;
29 import java.util.Map;
30 import java.util.Set;
31
32 public class Collection11 {
33     public static void main(String[] args) {
34
35         Hashtable<String, Integer> ht1 = new Hashtable();
36         Hashtable<String, Integer> ht2 = new Hashtable(15);
37         Hashtable<String, Integer> ht3 = new Hashtable(15, 0.60f);
38
39         // 1. public void put(Object key, Object value):
40         //     public Object putIfAbsent(Object key, Object Value):
41
42
43         ht1.put("CVM", 123);
44         ht1.put("VYP", 789);
45         ht1.put("PAT", 234);
46         ht1.put("HDS", 345);
47         ht1.put("MHM", 456);
48         ht1.put("CVM", 12);
49         System.out.println(ht1.putIfAbsent("CVM", 121));
50         System.out.println(ht1.putIfAbsent("ABC", 121));
51
52         System.out.println("ht1 is: "+ht1);
53

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54      // 2. public void putAll(Map m):
55
56      Hashtable<String, Integer> ht4 = new Hashtable(ht1);
57      ht4.putAll(ht1);
58      System.out.println(ht4);
59
60      // 3. public Object get(Object key):
61
62      System.out.println(ht1.get("CVM"));
63
64      // 4. public Object remove(Object key):
65      //      public Object remove (Object key, Object value):
66
67      System.out.println(ht1.remove("MHM", 456));
68      System.out.println(ht1.remove("CVM"));
69
70      System.out.println(ht1);
71
72      // 5. public int size():
73
74      System.out.println(ht1.size());
75
76      // 6. public boolean containsKey(Object key):
77
78      System.out.println(ht1.containsKey("PAT"));
79      System.out.println(ht1.containsKey("MHM"));
80
81      // 7. public boolean cotainsValue(Object key):
82
83      System.out.println(ht1.containsValue(234));
84      System.out.println(ht1.containsValue(123));
85
86      // 8. public Set keySet():
87
88      Set s = ht1.keySet();
89      System.out.println(s);
90
91      // 9. public Collection values():
92
93      Collection c = ht1.values();
94      System.out.println(c);
95
96      // 10. public boolean isEmpty():
97
98      System.out.println(ht1.isEmpty());
99      System.out.println(ht2.isEmpty());
100
101      // 11. Map.Entry:
102
103      for (Map.Entry m : ht1.entrySet()) {
104          System.out.println(m.getKey() + " = " + m.getValue());
105      }
106  }
107 }

```